Visual- and audio-aligned text encoders differentially predict EEG for auditory and visual words

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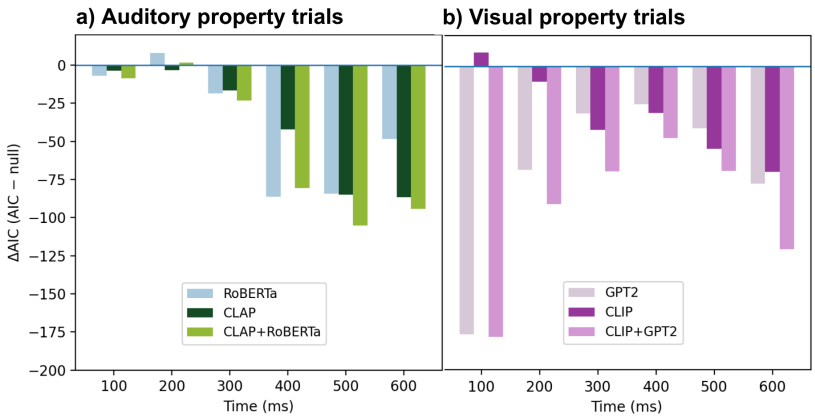
Background: In recent years, representing meanings as vectors learned from large language corpora has become a valuable approach for studying language in both minds and machines [4]. The vector embeddings make it possible to operationalize semantic contributions by combining different sources of information. Here we use embeddings from text encoders trained to align language with perceptual modalities, i.e., CLIP for vision and CLAP for audition, to approximate distributional representations informed with modality-specific perceptual information. We also use embeddings from text-only language models that are architecturally similar to the CLIP and CLAP text encoders (GPT2 and RoBERTa, respectively) to operationalize purely distributional semantics. We then test how well each model predicts EEG responses to visual and auditory property words in a property-verification task. The work bears on a key debate in lexical semantics: whether understanding word meaning depends on abstract, amodal structures, on representations shaped by perceptual experiences, or both. If human semantic representations primarily reflect distributional information, EEG elicited by words should be well predicted by text-only models. If grounded perceptual information also contributes, predictions should improve when using CLIP for visual property words and CLAP for auditory property words.

Method: We analyzed single-trial EEG data from 19 L1 English speakers performing a property-verification task, in which participants indicated whether a property term (e.g., “red”) is typical for a concept term (e.g., “APPLE”). Participants verified 192 experimental pairs involving visual properties (e.g., “red”) and auditory properties (e.g., “loud”). This analysis included 1,373 visual TRUE trials (e.g., APPLE-red) and 1,160 auditory TRUE trials (e.g., TRAINS-whistle), after excluding epochs with artifacts. Semantic dissimilarity was measured as the cosine distance between concept and property vectors from the final layers of GPT2 [5], RoBERTa [3], CLIP [6], and CLAP [8]. Logarithmic word frequency [2] served as a control predictor, and all predictors were z-scored. Mean voltage was computed at each electrode in successive 100-ms windows, from property-word onset to 700 ms. For each window, and separately for auditory and visual trials, we fit **ten** linear mixed-effects models, including single predictor, and pairing text models (GPT2, RoBERTa) with both multimodal models (CLIP, CLAP) to ensure that observed effects were not attributable to properties of a specific text encoder. The **NULL** model included no lexical or semantic predictor, but only the scalp-dimensions [7] as fixed effects, and item and subject-level random effects [voltage ~ (X+Y+Z) + (1|word) + (1|subject)]. The **WF** model added word frequency. Subsequent models added a single semantic predictor (**GPT2, CLIP**, **RoBERTa**, or **CLAP** cosine distance) or paired predictors (**CLIP+GPT2**; **CLAP+GPT2; CLIP+RoBERTa ; CLAP+RoBERTa** to the WF model. Models were compared using AIC, with $\Delta AIC \leq -10$ taken as robust evidence favoring the more complex model.

Results: For auditory trials (Fig. 1a), model fits showed little improvement over the NULL model before 300 ms. From 400 ms onward, CLAP and CLAP + RoBERTa produced progressively larger ΔAIC improvements, with both showing unique contributions from 500-700 ms. For visual trials (Fig. 1b), GPT2 outperformed the NULL in 0-100 ms and CLIP+GPT2 provided a robustly better fit than GPT2 from 200-700 ms. Among text-only models, GPT2 improved fits early in visual but not in auditory, whereas RoBERTa showed superior fits at 300-400 ms in visual trials and 300-500 ms in auditory trials (Fig 2).

Discussion: Embeddings from different models aligned with neural responses over time, and differed for auditory and visual property words, suggesting distinct representational features may become relevant for semantic processing depending on processing stages and contextual demands. Visual trials showed early sensitivity to GPT2’s prediction-driven embeddings, which may be related to low-level lexical predictability [1]. However, this effect was not observed for auditory words. RoBERTa showed superior fits in the 300-400 ms (viz. N400), suggesting that contextualized representations aligned better with neural activity. From 400 ms onwards, the vision-trained CLIP (controlling for GPT2), provided the strongest fit for the visual trials, and from 500 ms the audition-trained CLAP (controlling for RoBERTa) provided the best fit for auditory trials. This suggests unique contributions from both perception-linked and statistical regularities of the words, and may reflect task-driven processing dynamics in the later parts of the epoch. While broader training differences may also play a role in the observed dynamics, results suggest human sensitivity to perceptual similarity is not readily available from the statistics of language.

Figures:



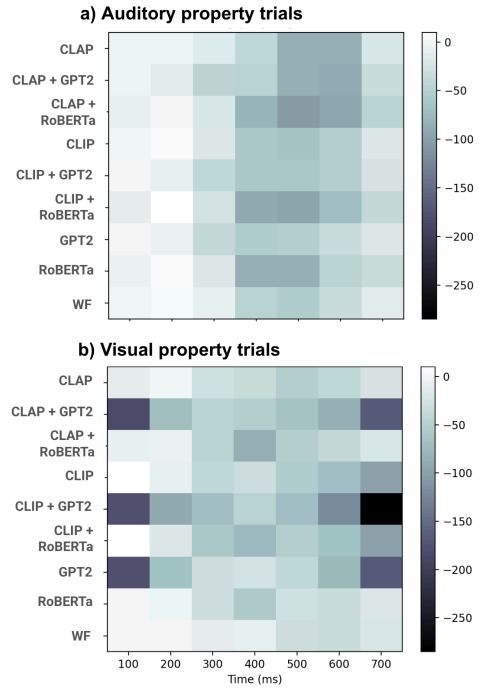


Figure 1: Multimodal model advantages for auditory (CLAP, 1a) and visual (CLIP, 1b) property trials in the first six 100ms windows. The final time window is omitted from the figure as the ΔAIC values are substantially larger in magnitude (≈ −270), which would compress variation in earlier windows and reduce interpretability.

Figure 2: Heatmaps showing ΔAIC values relative to the NULL model across seven 100-ms time windows for auditory (top) and visual (bottom) property trials. Rows correspond to models and columns to time.

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